

The Slope of a Tangent (1.2)



Today's Target

"I understand the slope of the tangent line to a curve as a limiting value. I can determine this limiting value algebraically. I can apply what I have learned in familiar and unfamiliar settings."

A1. Rates of Change



Class code



Close dialo

yjzjfwbc

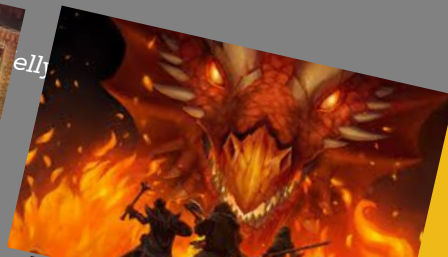
MEET ME IN A MINUTE!



HELLO!

My name is Ms. Kelly

I am here to support you with your learning in math.

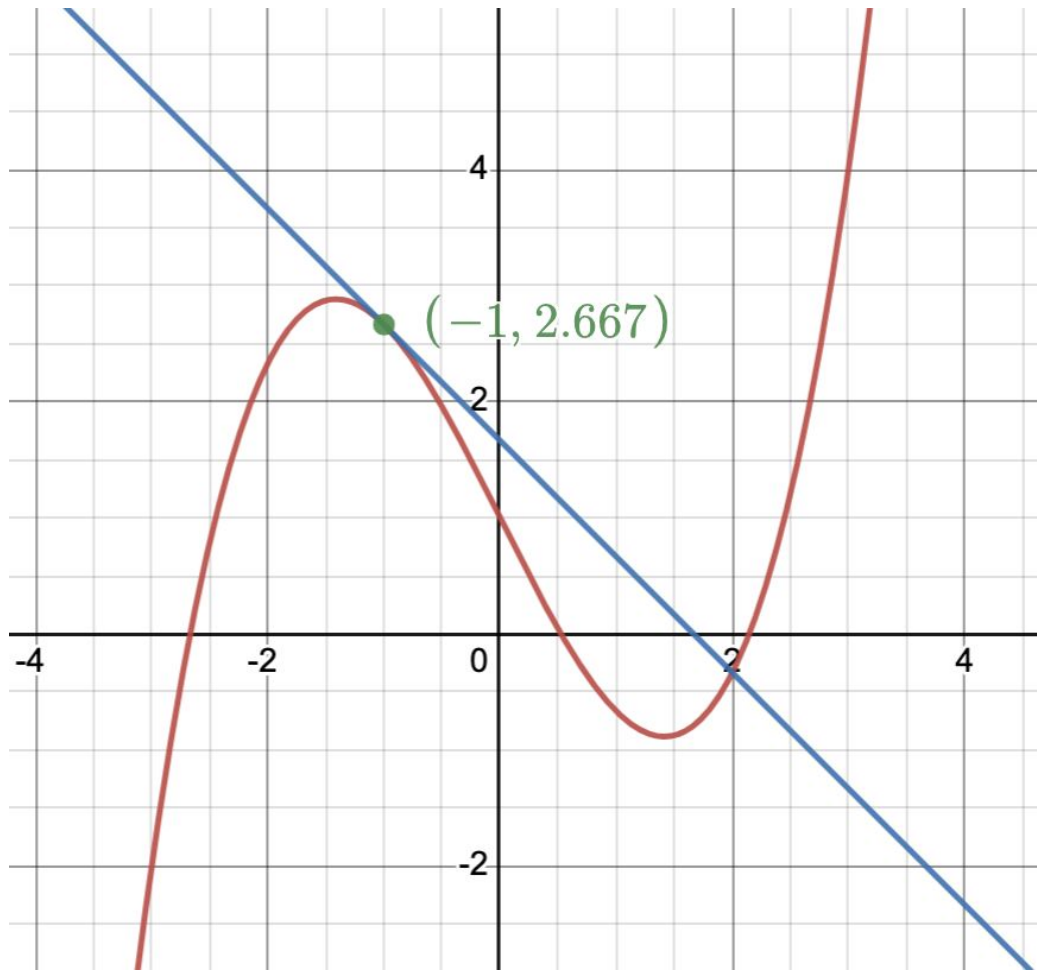


Observation:

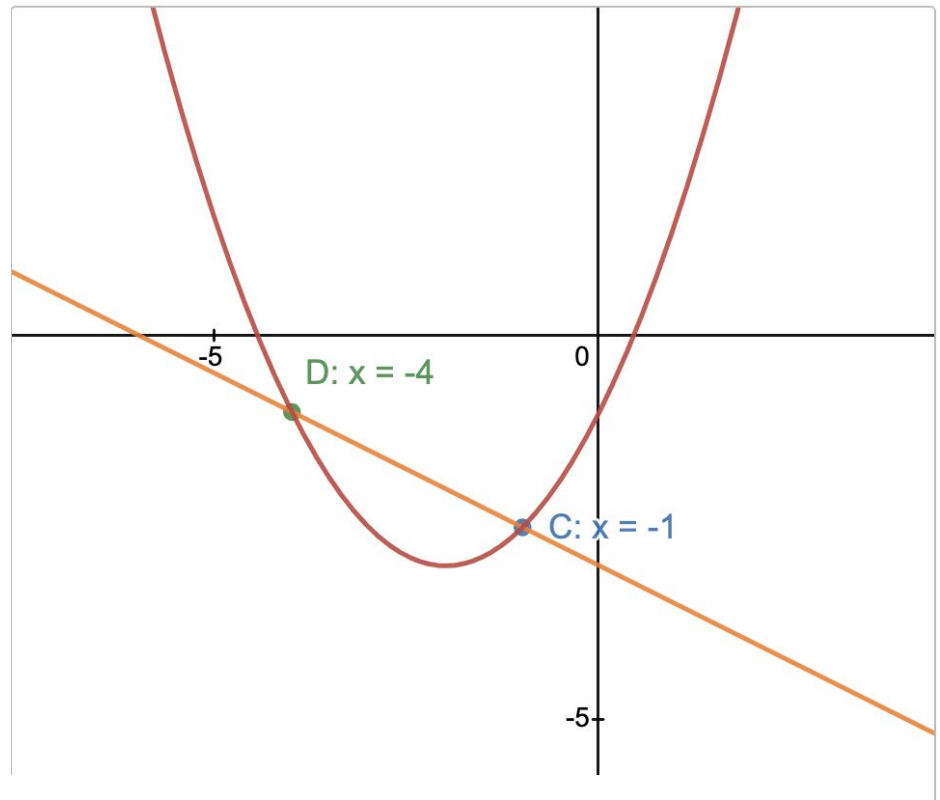
Point of Tangency- desmos

Follow the link in the
title:

ZOOM IN: What do you
notice as you zoom in
closer and closer to the
point of tangency
(green point)?



With your groups:
Secant line refresher



Find the slope of the orange secant line between the two points shown, given that $g(x)$ has equation $g(x) = \frac{1}{2}x^2 + 2x - 1$.

$$x = -4$$

$$g(x) = \frac{1}{2}x^2 + 2x - 1$$

$$AOC = \frac{\Delta y}{\Delta x}$$

$$AOC = \frac{g(-1) - g(-4)}{(-1) - (-4)} = \frac{8}{3}$$

$$g(-1) = \frac{1}{2}(-1)^2 + 2(-1) - 1$$

$$= \frac{1}{2} - 3$$

$$= -\frac{5}{2}$$

$$m = \frac{-1 + \frac{5}{2}}{-4 + 1}$$

$$m = \frac{\frac{3}{2}}{-3}$$

$$m = -\frac{1}{2}$$

$$AOC = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

$$= \frac{f(-1) - f(-4)}{-1 - (-4)}$$

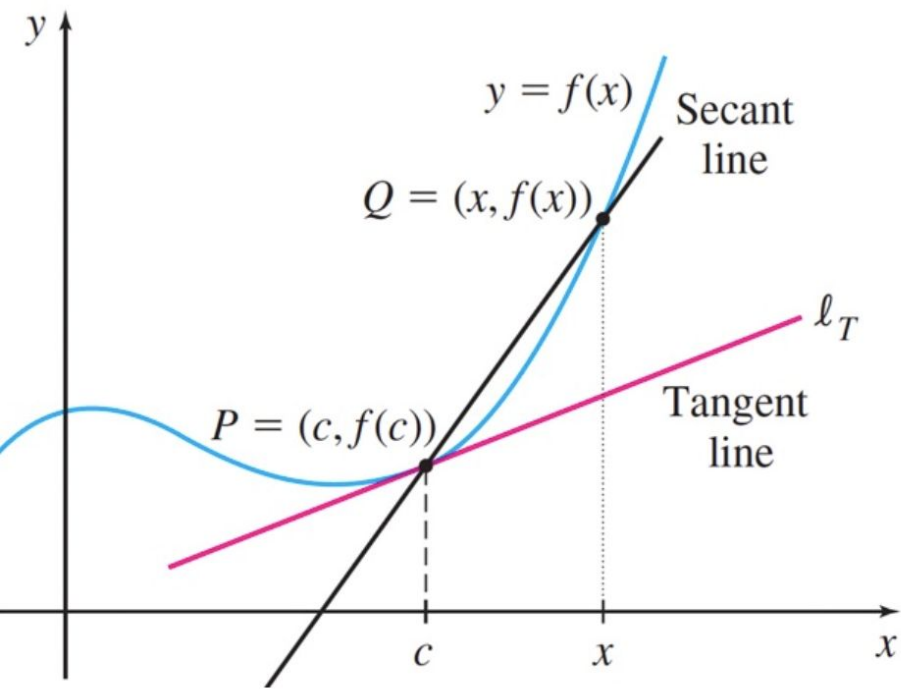
$$= \frac{-2.5 - (-1)}{3}$$

$$= \frac{-1.5}{3}$$

$$= -\frac{1}{2}$$

Information:

Secant lines vs Tangent lines



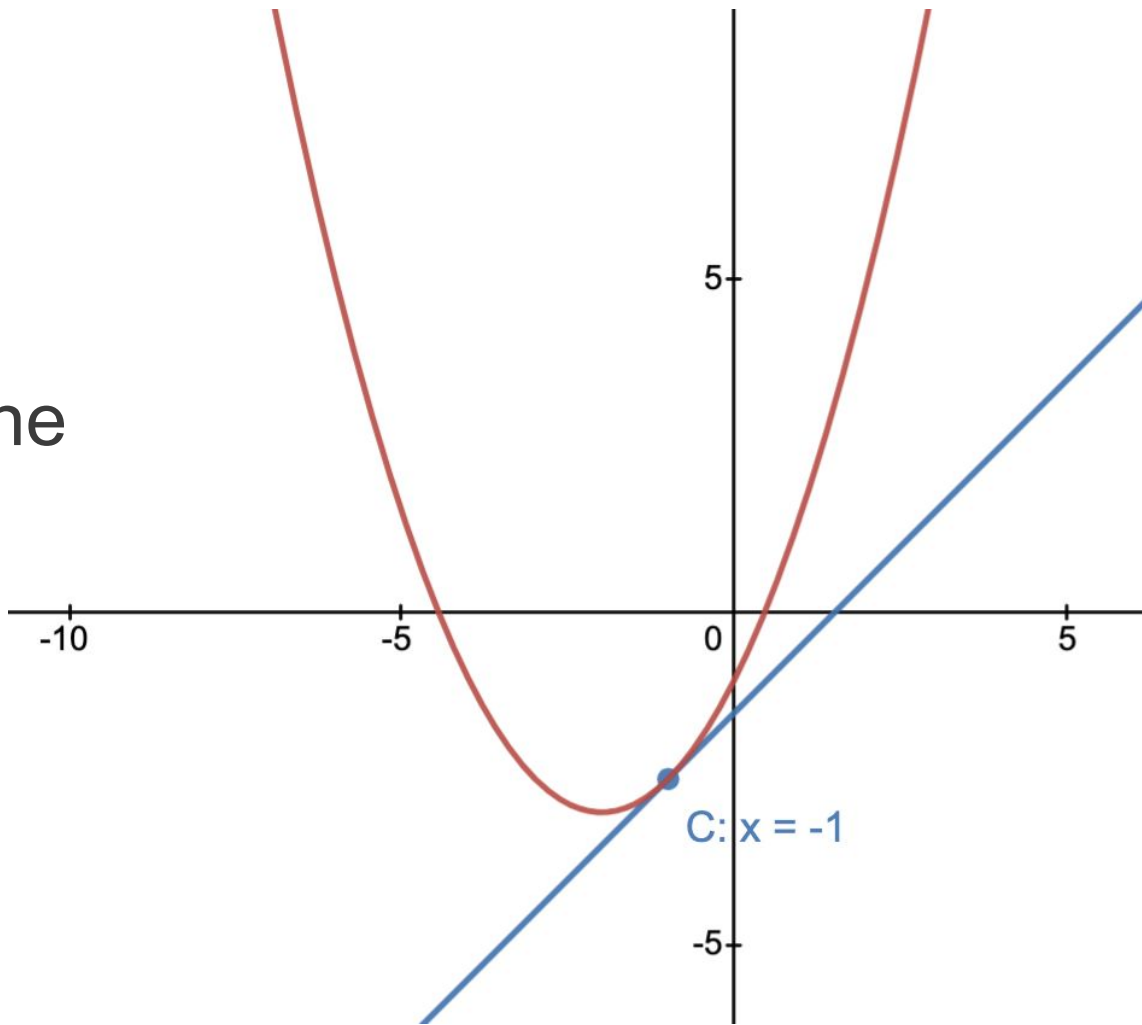
A **secant line** is a line joining two points on a function. The black line connecting points P & Q is a secant line.

The red line is a **tangent line** to $f(x)$ at point P, because it touches it at that single point locally and resembles the graph of the function at the point.

Discuss: Why is a tangent harder to find than a secant?

Slope of a **secant** line
~ slope of any line.

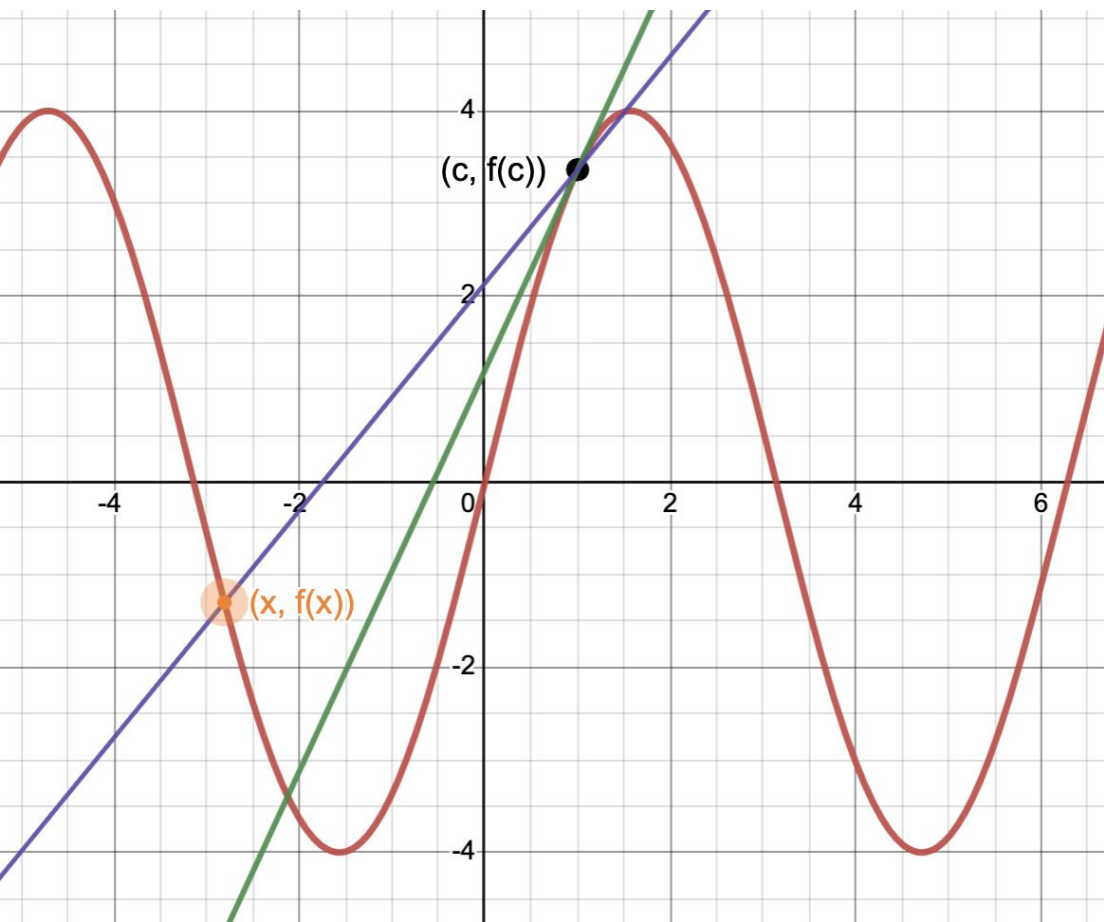
Slope of a **tangent**
line is a little
different...



Discover with your groups

[Student Desmos](#)

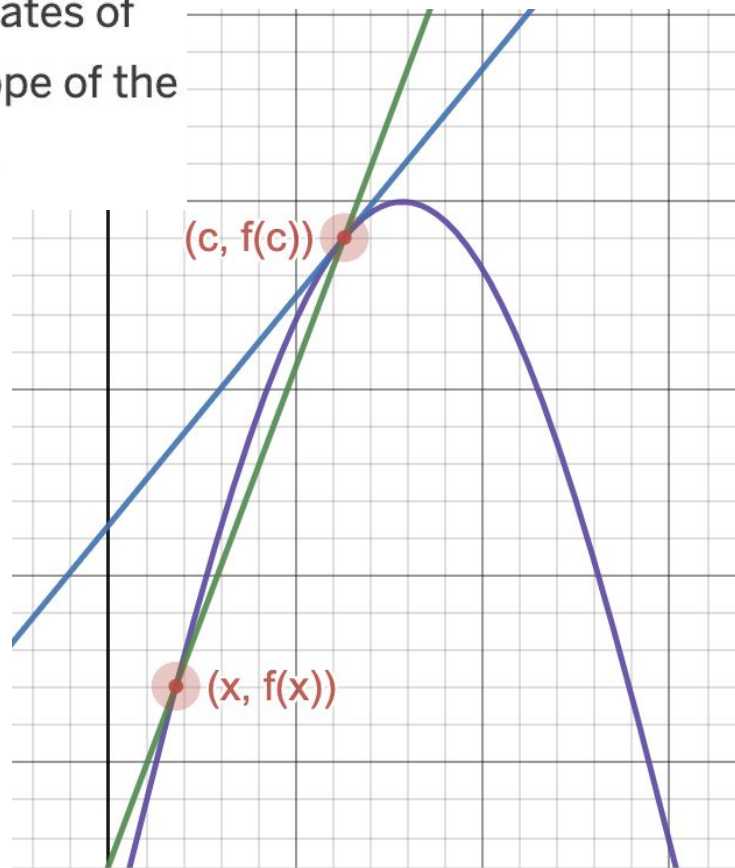
Demonstration: [Desmos](#)



The graph shows a green tangent line, a purple tangent line, and their slopes. Move an orange point toward the other red point. What do you notice?

Discuss

Use what you know about slope and the labeled coordinates of the points to write an expression that represents the slope of the secant line passing through $(c, f(c))$ and $(x, f(x))$.



Information: Secant

$$\frac{f(x) - f(c)}{x - c} \quad \text{or} \quad \frac{f(c) - f(x)}{c - x}$$

Information

If we let two points on the secant line get *VERY, VERY, VERY, close together*, the slope of the secant line is almost equal to the slope of the tangent line.

We need some notation for this idea of letting the points on the secant line get CLOSER and CLOSER and CLOSER together.

We're going to use something called a '**limit.**'

Information: Definition of a tangent

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

This means we are letting the point $(x, f(x))$ get as close as possible to having the same coordinates as $(c, f(c))$ without actually becoming the same point.

Discuss with your groups

What would happen if you tried to evaluate $\frac{f(x) - f(c)}{x - c}$ when $(c, f(c))$ and $(x, f(x))$ were the same point?

On paper- individually

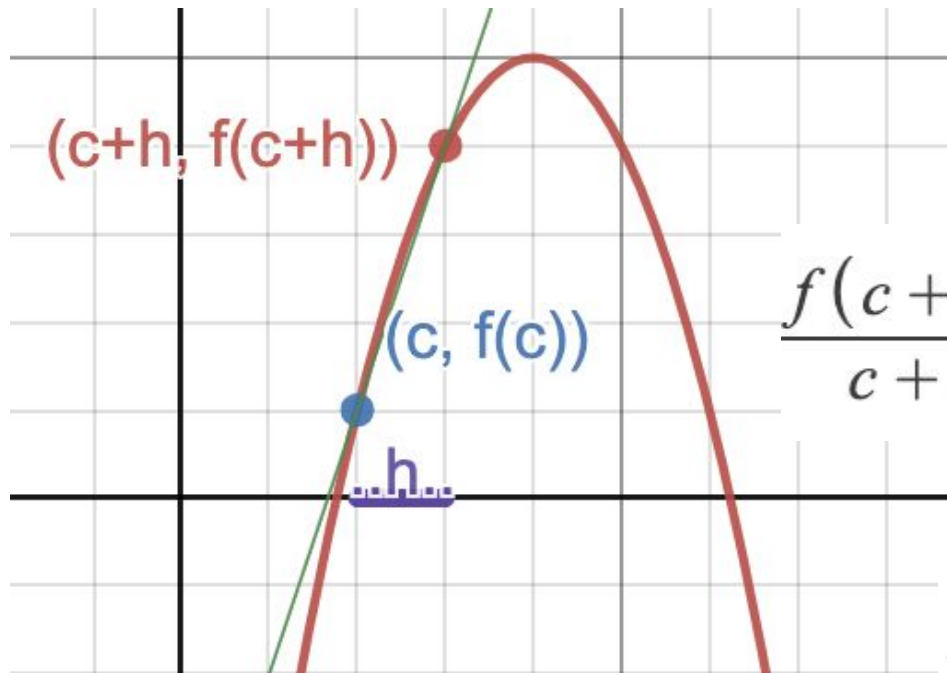
In your own words, explain why this expression represents the slope of the tangent to f at point $(c, f(c))$.

$$\lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c}$$

Information: Is there is another way?

(

let "h" represent the horizontal displacement from the blue point to the red point



Secant

$$\frac{f(c+h)-f(c)}{c+h-c} \quad \text{or} \quad \frac{f(c+h)-f(c)}{h}$$

Tangent

$$\lim_{h \rightarrow 0} \frac{f(c+h)-f(c)}{h}$$

Example

There is no "limit" to what you can accomplish.

Page 18 #	3a*	6a	9b	10a	11e	16	18	23	24
<i>Level</i>	1	1	3	3	3	3	2	4	4

* The final answer for 1.2 #3a may be expressed in slope/y-intercept form. It would be $y = (7/17)x - 40/17$